

Game Theory Elective

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0 Preamble: About this class

Remark 0.1 (Background and motivation of this course)

With these classes, I want to achieve at least four goals:

1. Most immediately, it should provide a **gentle introduction to game theory**. Students will learn how to interpret real decision problems as a *game*. Then they will learn how to identify solutions to these decision problems (e.g., to compute the game's *equilibria*), and to simulate learning in games.
2. More generally, the course should serve as an **example of mathematical modeling**. Students should have in mind that the framework of game theory (the formalism, the solution concepts) are human constructs. They should understand why smart people at some point found it useful to come up with this formalism; they should be able to critically question the underlying assumptions and methods. Ideally, when taking this course, students should put themselves into the shoes of a smart social scientist, a hundred years ago (when game theory was not yet an established formalism). Ideally, they try to come up with some of the formalism and some of the solution concepts themselves.
3. I hope the course will also just serve as **brainfood**. Students should enjoy to hear more about some scientific discipline that they might find stimulating, but that they would not hear about otherwise.
4. It should give you some impression of the research we do in our research group; this might become relevant once you decide on your **master thesis**.

Remark 0.2 (Prerequisites)

A fully rigorous game theory course requires quite a bit of math (e.g., matrices, derivatives, fixed point theorems). However, within this course we will focus on intuitions. We introduce some key concepts without much mathematical formalism. For students who are interested in seeing more details and math, I can suggest some relevant reading material (e.g. Fudenberg and Tirole, 1998, Sigmund, 2010).

Remark 0.3 (Some administrative remarks)

- Overall structure. The course consists of eight lectures. These lectures will tell you about what game theory is about, how one can translate real-world interactions into games, and how to 'solve' these games (where 'solving' could be interpreted as: coming up with a prediction of how certain people might play this game). As you will see, the specific solution might depend on what type of game is played, and what assumptions we make on individuals' rationality (more on that later).
- How to "pass" this part of the course. There will be in-class exercises (as a group), online assignments (individually), and a final conversation/interview (for which I will post a list of possible questions).

1 An overview on game theory

Remark 1.1 (What is game theory?)

Game theory is a scientific field that explores strategic decision-making in interactions with multiple decision-makers who jointly determine each others' outcomes. In particular, not all decision-making is covered by game theory; as an example, the following questions are not covered:

How to play roulette? Should I buy a given flat at a given price? Should I do a PhD?

Instead, a strategic component is important. So, the following variations of the previous questions are in fact covered by game theory:

How to play poker? How to bargain the price of a flat you want to buy? How to apply to a PhD program?

In the following, we will ask two questions: (1) How can we formalize such strategic interactions in a mathematical model? (2) Given we have a model, what does it mean to *solve* a game?

In-Class Exercise 1.2 (Some examples to form intuitions)

Are the following questions covered by game theory (and if so, why?): (1) *Should I actively participate in this course?* (2) *Should I be modest or bold?* (3) *What should I wear today?*

Example 1.3 (Guess 2/3 of the average)

Let's play the following game. Everyone secretly writes down a number between 0 and 100. Then we compute the average of all numbers. Whoever is closest to $2/3$ of this average, wins 2 EUR.

Remark 1.4 (Elements of a game)

When formalizing strategic interactions as a game, it is important to capture the following five elements:

- Players. Who are the relevant individuals that make (strategic) decisions?
- Actions. Between which actions can the players decide?
- Order of moves. Who gets to make a decision at which point?
- Information. What is it that players know when making their decisions?
- Payoffs. Depending on the players' decisions, what is the eventual outcome for each player? [e.g., measured in money, or happiness, or *utility*]

Example 1.5 (Revisiting auctions)

Suppose there is a piece of art by a famous artist that you might want to buy. You know the value that this piece of art has to you.. You don't know the value that other potential buyers assign to it. But you know how many other buyers there are, and you know that for each of them, their evaluation of the piece of art is randomly drawn from $\{0 \text{ EUR}, 1 \text{ EUR}, 2 \text{ EUR}, \dots, 10 \text{ EUR}\}$. The rules of the auction are: Everyone secretly writes down the price they would be willing to pay (their *bid*). The potential buyer with the highest bid needs to pay the respective price, but gets the piece of art in return. (If there are multiple

buyers with the same bid, it's randomly allocated). So for example, if you happen to assign a value of 8 EUR to the piece of art, and if you have the highest bid (which happens to be 5 EUR), you will have a profit of 3 EUR. For this example of an auction, how would we define each of these five elements?

- *Players*. All the potential bidders, $\mathcal{N} = \{1, 2, \dots, n\}$.
- *Actions*. Each player needs to name a price they are willing to pay, $p_i \in \{0, 1, \dots, 10\}$ for $i \in \mathcal{N}$.
- *Order of moves*. All players need to make their decision simultaneously.
- *Information*. Players know how their own private value of the good, $v_i \in \{0, 1, \dots, 10\}$. They don't know the precise value of the other players, but they know it's uniformly distributed in $\{0, 1, \dots, 10\}$.
- *Payoffs*. The player who gets the item has a payoff of $v_i - p_i$; all others have a payoff of zero.

In-Class Exercise 1.6 (Split or Steal)

Split or Steal used to be the final stage of the UK game show *Golden Balls*. It involves two participants, each having two balls, marked 'Split' and 'Steal'. Participants independently choose one of the balls. If they both split, they share the prize money. If one splits and the other steals, the latter gets all the prize money. If they both steal, no one gets anything. Prior to making a decision, participants are allowed to talk to each other. Some examples:

<https://www.youtube.com/watch?v=bJzsYLSPY7o>

<https://www.youtube.com/watch?v=S0qjK3TWZE8&t=159s>

How would you define each of the five elements of this game?

Remark 1.7 (An overview on different kinds of game theories)

There are several different approaches to game theory:

1. Classical/Standard game theory. This kind of game theory asks what rational players would do. Being rational here means that players completely understand all aspects of the game, and that they wish to maximize their payoffs.
2. Evolutionary game theory. Here, players do not need to be rational. Rather they are myopic, but they try to improve their payoffs over time.
3. Epistemic game theory. This kind of game theory tries to understand how different assumptions on what people know shapes the predictions of game theory (e.g., if players are no longer assumed to be rational, or if players no longer believe in their co-players' rationality).
4. Algorithmic game theory. Here, researchers are concerned with the computational complexity of solving games, and in identifying optimal algorithms for doing so.
5. Experimental game theory. Here the question is: How do actual people behave in strategic interactions?

We will mostly focus on classical game theory, but we'll also have a brief look at evolutionary and experimental game theory (that's also what we do in our research group).

Remark 1.8 (Different classes of games)

There are various ways how to classify games. One common distinction is whether all players decide

once and simultaneously, or whether they make (possibly multiple) decisions over time (think of chess, for example). The respective game classes are called *static games* and *dynamic games*, respectively. Another distinction is whether players know all relevant aspects of the game. Here, the distinction is between *games with complete information* and *games with incomplete information* (think of chess vs poker).

2 Static games with complete information

2.1 Normal-form games

Remark 2.1 (Normal-form games)

In the following, we start by discussing the simplest possible case: static games with complete information among two players (who can choose among finitely many actions). That is, players make only one decision, and they need to decide simultaneously. When making their decision, they know all relevant aspects of the game (e.g., the possible actions and the feasible payoffs). The only thing they do not know is the decision of the other players. These games are called *normal form games*. They can be represented by a table (*payoff matrix*).

Example 2.2 (Split or Steal)

Let's revisit the game 'Split or Steal'. There are two players, who can choose among two actions. Depending on the players' actions, they either share the prize P , one of the players gets all, or they both get nothing. Ignoring the pre-game communication state, the situation can be represented by the table

	Split	Steal
Split	$P/2, P/2$	$0, P$
Steal	$P, 0$	$0, 0$

Here, one of the players (the blue player) chooses a row of the matrix and the other (the red player) chooses a column. The entries of the table represent the payoff of the respective player. Players wish to maximize their realized payoff.

In-Class Exercise 2.3 (Deriving payoff matrices: Snowdrift game, stag-hunt game)

1. There is a task that needs to be done (e.g., washing the dirty dishes in the kitchen sink). There are two individuals who need to decide whether or not to volunteer for doing the task. Both individuals prefer that the task is done, but they both prefer the other one to do it. Assume if at least one person volunteers, both get a benefit b . However, anyone who volunteers pays some small cost $c < b$. If both players volunteer, they share the cost.
2. There are two members of a tribe who need to decide whether to hunt stag or hare. To successfully hunt a stag, it takes two committed hunters. In that case, each hunter gets a payoff of 10. All by yourself, you cannot get a stag and you would get a payoff of 0 if you tried. Hunting a hare can be done individually, without any risks, leading to a safe (but smaller) payoff of 4, independent of the other tribe member.

Remark 2.4 (Solving games)

So far, we have encountered several instances of strategic interactions, and we have translated them into *games*. Now that we have this game representation, we ask ourselves: How can we predict what *rational* individuals would do in such games? Or to put differently, how would those individuals *solve* these games? In the following, we discuss two solution concepts: *elimination of dominated actions* and the *Nash equilibrium*.

2.2 Elimination of dominated actions**Remark 2.5** (An intuitive solution for (a modified) Split and Steal)

Consider a Split and Steal game with a prize of 10,000 EUR. Moreover, suppose players feel really unhappy when they split while the opponent steals (represented by an emotional cost of 100 EUR). Then the resulting game can be represented by

	Split	Steal
Split	5,000, 5,000	-100, 10,000
Steal	10,000, -100	0, 0

In this game, it is noteworthy that while splitting might seem to be the social choice, we probably would predict that rational players would steal. The reason is: even if I'm unsure about what the other player is doing, by stealing I am always better off. We can represent this observation by drawing arrows into the payoff matrix. For each action of the opponent, these arrows indicate which action the focal player would rather choose:

	Split		Steal
Split	5,000, 5,000	→	-100, 10,000
	↓		↓
Steal	10,000, -100	→	0, 0

This representation reinforces our earlier observation: individually, each player prefers stealing, *no matter what the other player would do*.

Definition 2.6 (Dominated actions)

An action a is (strictly) dominated if there is some other action a' which always yields a larger payoff.

Remark 2.7 (On dominated actions and rationality)

The notion of dominated actions captures some kind of minimum requirement that we would expect rational players to do: They should never use actions that are never optimal, for no possible belief on what their co-player might be doing. Hence, we should expect rational players to *eliminate* dominated actions. If, after elimination of all dominated actions, there is only one action per player left, we may call the remaining strategy pair a solution of the game. In this sense, the above 'Split and Steal' can be solved, and the solution is for both players to steal (even though that outcome leaves everyone worse off on a collective level).

In-Class Exercise 2.8 (A model of teamwork)

Let us consider a version of the so-called *public goods game*. In this game, there is a group of n players. Each player can either contribute to some collective effort or free-ride. A contributing player pays a cost c , which generates a benefit of $r \cdot c$ that is evenly divided among all group members. A free rider pays nothing, generates no benefit, but still gets their share from the benefits generated by others. As an example, consider a group with $n=3$ players, a cost of $c=3$, and a multiplication factor of $r=2$. We can represent this scenario with two payoff matrices. Player 1 chooses a row, player 2 chooses a column, and player 3 chooses one of the two payoff matrices,

		Contribute		Free-ride	
		Contribute	Free-ride	Contribute	Free-ride
Contribute		3, 3, 3	1, 4, 1	1, 1, 4	-1, 2, 2
Free-ride		4, 1, 1	2, 2, -1	2, -1, 2	0, 0, 0

In this game, contributing is a dominated action. For any pair of actions that the other two players may choose, the remaining player is always better off free riding. This logic suggests that in the end everyone will free-ride (giving a payoff of zero to everyone), even though they all would benefit from collectively contributing (payoff of three to everyone). Models like these are used to explain the nations' reluctance to take measures against climate change, or the over-use of public resources (Hardin, 1968).

Homework Exercise 2.9 (Eliminating dominated actions)

Being able to identify dominated actions is important; so here is an exercise to practice this skill. Suppose there are two players. Player 1 can choose between the three actions T, C, B (top, center, bottom), whereas player 2 can choose between L, M, R (left, middle, right). Payoffs are given by the matrix

	L	M	R
T	2, 3	5, 0	1, 4
C	0, 3	8, 1	3, 2
B	1, 9	3, 7	0, 8

Does either player have one (or several) dominated actions? Remember, to show whether some action a for player 1 is dominated, you need to find some other action a' that has the property: for any action that the co-player might choose, a' gives a larger payoff to player 1 than a (so even if player 1 does not know what the co-player will do, action a' is always a better choice than a). The way to identify dominated actions for player 2 is similar. Find all dominated actions and justify why they are dominated. Also justify why none of the remaining actions is dominated.

Remark 2.10 (A Split and Steal game with remorse)

Unfortunately, not all games can be solved by eliminating dominated actions. Consider a version of the above Split and Steal where the (blue) row-player would have a bad conscience when stealing against a splitting opponent. In that outcome, the row-player would experience some emotional cost of 8,000 EUR.

Then the resulting payoff matrix is

	Split	Steal
Split	5,000, 5,000	-100, 10,000
Steal	2,000, -100	0, 0

Now, the row-player is actually open to splitting the amount (provided the co-player goes along). Nevertheless, if both players can be assumed to be rational, there are still reasons to believe they would both steal eventually. The reason is: For the (red) column-player, splitting is still a dominated action; hence that player will certainly prefer to steal. However, if the row-player is aware of its opponent's rationality, the row-player will anticipate that the opponent will certainly steal. Given that belief, it also becomes better for the row-player to steal. This is how the dynamics can be represented:

	STEP 1	
	Split	Steal
Split	5,000, 5,000	-100, 10,000
Steal	2,000, -100	0, 0

Because the red column-player is rational, this player will eliminate the Split option ('Step 1'). Because the blue row-player knows of the opponent's rationality, this row-player will respond by also eliminating the Split option ('Step 2'). In the end, both players steal.

Remark 2.11 (Iterated elimination of dominated actions)

- The above procedure to arrive at a prediction is called iterated elimination of dominated actions. Here, we first eliminate actions that are dominated for either of the two players. In a next step, we consider the reduced game that only contains actions that haven't been eliminated yet. Then we also eliminate all actions that are dominated in this reduced game. We iterate this process until no action can be eliminated anymore. If in the end, both players only have a single action, then the game is solvable by iterated elimination of dominated actions (such games are also called 'dominance-solvable').
- The above Split and Steal with Remorse is dominance-solvable. However, we note that the required reasoning on part of the players is now more complex. To arrive at this solution, not only does the row-player need to be rational (try to maximize her own payoff). She also needs to believe that her opponent is rational, and she needs to use that belief to correctly anticipate the opponent's behavior.

Example 2.12 (Cournot duopoly)

There are two firms who both produce the same product, and hence compete in the same market. Ahead of the next season, each firm needs to decide how many quantities x of their product to produce. For simplicity, we assume they can only choose between three quantities, $x \in \{10, 20, 30\}$. The production cost of one item is $c=1$. The more products there are, the lower the market price p that the two companies can ask for,

$$p = 70 - (x_1 + x_2). \quad (1)$$

Companies aim to maximize their profit, which is given by their revenue minus their costs. For example,

for firm 1 this yields the payoff function

$$\pi_1 = (p - c)x_1 = (70 - (x_1 + x_2) - 1)x_1.$$

This scenario can be represented with the following payoff matrix,

	$x_2 = 10$	$x_2 = 20$	$x_2 = 30$
$x_1 = 10$	490, 490	390, 780	290, 870
$x_1 = 20$	780, 390	580, 580	380, 570
$x_1 = 30$	870, 290	570, 380	270, 270

We can solve this game by iterated elimination of dominated actions. As our first step, we note that producing $x_i = 10$ is dominated by producing $x_i = 20$: depending on the other firm's actions, the profits would be 780 (instead of 490), 580 (instead of 390), or 380 (instead of 290). However, once both firms realize that no firm would ever produce $x_i = 10$, we have a reduced game with two actions, $x_i \in \{20, 30\}$. In that reduced game (but not in the original game), producing $x_i = 30$ is dominated by $x_i = 20$. Here, profits would be 570 (instead of 580), or 270 (instead of 380). Therefore, iterated elimination of dominated actions suggests that both firms will produce $x_i = 20$. The resulting price according to Eq. (1) is $p = 30$.

In-Class Exercise 2.13 (Traveler's dilemma)

Consider the following scenario by Basu (1994): An airline lost two identical pieces of luggage by two different travelers. It makes the following offer: It will ask both travelers independently the value of the luggage (within a range between 180 EUR and 300 EUR). Then it will pay the lower of the two claims. But in case the two claims are different, it will also pay a small reward of 5 EUR to the traveler who made the smaller claim. In this game, who are the players, what are their possible actions, and what are their possible payoffs? Can we solve this game by elimination of dominated actions? Can we solve it by iterated elimination of dominated actions?

Optional Exercise 2.14 (Guess 2/3 of the average, revisited)

In our first class, everyone wrote down a number between 0 and 100. Then, whoever was closest to 2/3 of the average won the jackpot (2 EUR). By a similar logic as in the traveler's dilemma, can you solve this game by iterated elimination of dominated actions?

Remark 2.15 (Connection to the other game theories)

1. Epistemic game theory. Epistemic game theory explores how rational individuals need to be, and how strongly they need to believe in their co-player's rationality, for certain solution concepts to make sense. For the traveler's dilemma, we can come up with a unique solution if we accept that each player needs to be rational, needs to be aware of its co-player's rationality, needs to be aware that the co-player is aware of one's own rationality, etc. That seems quite restrictive!
2. Experimental game theory. Goeree and Holt (2001) have tested the traveler's dilemma experimentally. The two travelers need to claim a value between 180 and 300. There are two treatments, either with a small reward ($R = 5$) or a large reward ($R = 180$). The magnitude of the reward does not affect

the theoretical prediction by dominance solvability. However, it has huge effects on the experimental outcome. When the reward is large, most people ($\sim 80\%$) make the smallest possible claim of 180. When the reward is small, most people (again $\sim 80\%$) make the largest possible claim of 300.

3. Empirical observations on ‘Split or Steal’. ven den Assem et al. (2011) analyzed participant’s behavior in the game show. Among other findings, they report that on average participants split in roughly 53% of the instances. There is some limited support that cooperation rates decrease with increasing stakes. Also, men are somewhat more likely to steal, but old men are not.

Remark 2.16 (Conclusion: On elimination of dominated actions)

After having introduced normal form games, in this subsection we described an elementary method to *solve* such games. This method is based on the insight that rational players would not want to use obviously inferior actions (and they would expect their co-players not to use such actions either). Based on this method, we could *solve* a number of games, such as Split-or-Steal, the public good game, or the traveler’s dilemma. However, it is fairly easy to come up with games that are not dominance-solvable. In the following, we look at one such game to come up with a more general solution concept.

2.3 Nash equilibrium

Remark 2.17 (Optimal behavior in the stag-hunt game)

One example of a game that has no dominated actions is the stag-hunt game with payoff matrix

	Stag	Hare
Stag	10, 10	0, 4
Hare	4, 0	4, 4

Opting for **Stag** is not dominated – after all, it is the best thing I can do if I expect my co-player to hunt stag as well. Similarly, hunting **Hare** is not dominated either. Still, some outcomes seem less likely than others. For example, it would be odd to consider (**Stag**, **Hare**) a ‘solution’ (i.e., for the row-player to go for a stag-hunt, while the column-player hunts a hare). Here, at least one of the players would immediately want to deviate from that outcome. In that regard, an outcome like (**Hare**, **Hare**) or (**Stag**, **Stag**) seems much more reasonable.

Definition 2.18 (Nash equilibrium)

A pair of actions (a_R, a_C) for the two players (one action for the row-player and one for the column-player) is a Nash equilibrium if neither player can improve its payoff by unilaterally deviating.

Example 2.19 (Stag-hunt, revisited)

There is a fairly simple algorithm to identify the Nash equilibrium action pairs. For each action of the opponent, draw a square around the focal player’s largest payoff. For example, in the stag-hunt game, if the opponent chooses **Stag**, the focal player’s best response is **Stag**, which we highlight by 10. By

repeating the same steps for all other cases, we obtain

	Stag	Hare
Stag	10, 10	0, 4
Hare	4, 0	4, 4

Any cell of the matrix that contains two squares is a Nash equilibrium. In this case, there are two such equilibria, (Stag, Stag) and (Hare, Hare).

In-Class Exercise 2.20 (Two examples)

1. What are the Nash equilibria of the snowdrift game (assuming the benefit of cooperation b is larger than the cost c of cooperating),

	Cooperate	Defect
Cooperate	$b - c/2, b - c/2$	$b - c, b$
Defect	$b, b - c$	$0, 0$

2. Penalty kicks. Consider a game between a striker and a goalkeeper. The striker needs to choose whether to shoot left or right, the goalkeeper chooses in which direction to jump. To simplify matters, suppose the striker scores if and only if the goalkeeper jumps into the wrong direction. Moreover, if we assume the striker is the row-player, we can represent this game as follows,

	Left	Right
Left	0, 1	1, 0
Right	1, 0	0, 1

Can we find a pair of actions that is a Nash equilibrium in that game?

Remark 2.21 (Mixed strategies)

The game of penalty kicks suggests that sometimes, players may wish to be unpredictable (there are many other such examples, such as in Rock-Scissors-Paper, or in warfare). We can incorporate this by saying that players do not choose a single action with certainty; rather they play their actions according to some probability distribution $\mathbf{x} = (x_L, x_R)$. In that case, we say that the player adopts a ‘mixed strategy’ (obviously, these numbers need to add up to one, $x_L + x_R = 1$. After all, each player is guaranteed to choose *some* action). If the mixed strategy specifies that one action should be played with certainty, e.g., $x_L = 1$ and $x_R = 0$, we also speak of a ‘pure strategy’.

Remark 2.22 (Calculating with mixed strategies)

1. **Computing payoffs.** One can define a payoff for mixed strategies by considering their expected value. For example, consider the penalty-kicks game in **Exercise 2.20**. Moreover, suppose the goalkeeper (in red) chooses a mixed strategy of jumping left in 40% of all cases and jumping right in 60% of all cases,

$\mathbf{y} = (y_L, y_R) = (0.4, 0.6)$. That is, the situation for the (blue) striker can be represented as follows,

	Left (40%)	Right (60%)
Left	0, 1	1, 0
Right	1, 0	0, 1

Then we can compute the payoff of a striker who always shoots left as

$$\pi_1(\text{Left}, \mathbf{y}) = 0.4 \cdot 0 + 0.6 \cdot 1 = 0.6$$

Similarly, the payoff for always shooting right is

$$\pi_1(\text{Right}, \mathbf{y}) = 0.4 \cdot 1 + 0.6 \cdot 0 = 0.4$$

If the striker now uses a mixed strategy too, $\mathbf{x} = (x_L, x_R) = (x, 1 - x)$ her payoff is just the weighted average of the two previous payoffs,

$$\pi_1(\mathbf{x}, \mathbf{y}) = x \cdot \pi_1(\text{Left}, \mathbf{y}) + (1 - x) \cdot \pi_1(\text{Right}, \mathbf{y}) = x \cdot 0.6 + (1 - x) \cdot 0.4 = 0.4 + 0.2 \cdot x.$$

Given the requirement $x \in [0, 1]$, this payoff function $\pi_1(\mathbf{x}, \mathbf{y})$ has a maximum at $x = 1$. That is, given that the goalkeeper is slightly biased to jump to the right, the striker's *best response* is to always shoot to the left.

2. **Defining Nash equilibria.** One can extend our earlier definition of Nash equilibria to the case of mixed strategies. A pair of strategies $\mathbf{x} = (x, 1 - x)$ and $\mathbf{y} = (y, 1 - y)$ is an equilibrium if neither player has an incentive to deviate. That is, the first player's payoff $\pi_1(\mathbf{x}, \mathbf{y})$ has a maximum at $\mathbf{x}$ and the second player's payoff $\pi_2(\mathbf{x}, \mathbf{y})$ has a maximum at $\mathbf{y}$.

Based on this definition, we can confirm the intuition that $\mathbf{x}^* = (0.5, 0.5)$ and $\mathbf{y}^* = (0.5, 0.5)$ is a Nash equilibrium. In fact, given the goalkeepers' strategy $\mathbf{y}^* = (0.5, 0.5)$, we can again compute the striker's payoff from shooting left and right, respectively,

$$\begin{aligned} \pi_1(\text{Left}, \mathbf{y}^*) &= 0.5 \cdot 0 + 0.5 \cdot 1 = 0.5, \\ \pi_1(\text{Right}, \mathbf{y}^*) &= 0.5 \cdot 1 + 0.5 \cdot 0 = 0.5. \end{aligned} \tag{2}$$

In particular, when the striker uses a mixed strategy $\mathbf{x} = (x, 1 - x)$, her payoff is

$$\pi_1(\mathbf{x}, \mathbf{y}) = x \cdot \pi_1(\text{Left}, \mathbf{y}) + (1 - x) \cdot \pi_1(\text{Right}, \mathbf{y}) = x \cdot 0.5 + (1 - x) \cdot 0.5 = 0.5.$$

That is, given the goalkeeper's strategy, the striker's payoff does not even depend on her probability x to shoot left. In particular, she has no incentive to deviate from choosing $x^* = 0.5$. By symmetry of the game, the same logic applies to the goalkeeper; given that the striker chooses each corner with equal probability, also the goalkeeper has no incentive to deviate from $y^* = 0.5$.

3. **Finding Nash equilibria in mixed strategies.** In the penalty-kick example, the Nash equilibrium

(x^*, y^*) had the following curious property: Given the goalkeeper's strategy, the striker got the same payoff from always choosing left and from always choosing right, see Eq. (2). In fact, for normal-form games, one can show that this property needs to hold for any Nash equilibrium in mixed strategies. In equilibrium, players always need to be indifferent between the options they choose (otherwise they would have an incentive to choose the more profitable option more often). This insight can be used as a recipe to find mixed Nash equilibria. For example, if we require player 1 to be indifferent between choosing left or right for a given strategy $y = (y, 1-y)$, this yields the equation

$$\pi_1(\text{Left}, y) = \pi_1(\text{Right}, y).$$

By using the respective payoff formulas, this equation is equivalent to

$$y \cdot 0 + (1-y) \cdot 1 = y \cdot 1 + (1-y) \cdot 0.$$

This equation has a unique solution, $y^* = 0.5$. Analogously, we can also confirm $x^* = 0.5$. Based on this insight, we are no longer dependent on intuitions to 'guess' the position of the mixed Nash equilibrium.

In-Class Exercise 2.23 (A more realistic version of penalty-kicks)

In the previous example, we assumed that the striker always scores when the goalkeeper chooses the wrong corner. We also assumed that the striker never scores when the goalkeeper chooses the correct corner. Finally, we assumed that the striker is equally good at shooting towards the left-hand side and the right-hand side. In fact, strikers have a better success rate when shooting towards their 'natural side' (for a right footed striker that would be shooting towards the left, Chiappori et al., 2002). Under these assumptions, a more realistic payoff matrix may look like this,

	Left	Right
Left	0.4, 0.6	0.9, 0.1
Right	0.8, 0.2	0.3, 0.7

What is the mixed Nash equilibrium in this case?

Optional Exercise 2.24 (The snowdrift game revisited)

In **Exercise 2.20**, we have already looked for those equilibria of the snowdrift game where both players choose a pure strategy (i.e., they choose a single action with certainty). This was the solution:

	Cooperate	Defect
Cooperate	$b - c/2, b - c/2$	$b - c, b$
Defect	$b, b - c$	$0, 0$

In addition, the game has a 'mixed' equilibrium in which both players cooperate with some probability $x^* = y^*$. Following the steps in **Remark 2.22(iii)**, can you find this probability?

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